

Google Ads API Tool Design Document

NetCo Group - Internal Marketing Dashboard

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Brands: NetCo Body-Cam | BauTV+ | Microvista

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1. Company Overview

NetCo Group is an association of three companies based in Germany:

- NetCo Body-Cam: Sales and rental of professional bodycam systems for security personnel, police, and private companies.
- BauTV+: Sales and rental of camera and monitoring systems for construction sites, including time-lapse documentation.
- Microvista: Industrial computed tomography (CT) services for non-destructive testing and quality assurance.

All three brands use Google Ads for lead generation across Search, Display, Video, and Performance Max campaigns. The company operates from Germany and serves customers across the DACH region and the EU.

2. Tool Overview

2.1 Purpose

We are building an internal marketing dashboard that consolidates performance data from multiple sources (Google Ads, Google Search Console, social media) into a single unified view. The Google Ads API integration is specifically used for READ-ONLY reporting purposes - importing campaign, ad group, and keyword performance metrics for visualization and analysis.

2.2 Tool Name

NetCo Marketing Dashboard

2.3 Access

The tool is strictly internal. Only employees of the NetCo Group have access to the dashboard. There are no external users, no client-facing features, and no plans to commercialize or distribute the tool. Access is controlled via authentication (Vercel deployment with restricted access).

2.4 Technology Stack

- Frontend: Next.js 14 (React) with shadcn/ui components
- Backend: Convex (real-time database and serverless functions)
- Hosting: Vercel (frontend), Convex Cloud (backend)
- Language: TypeScript

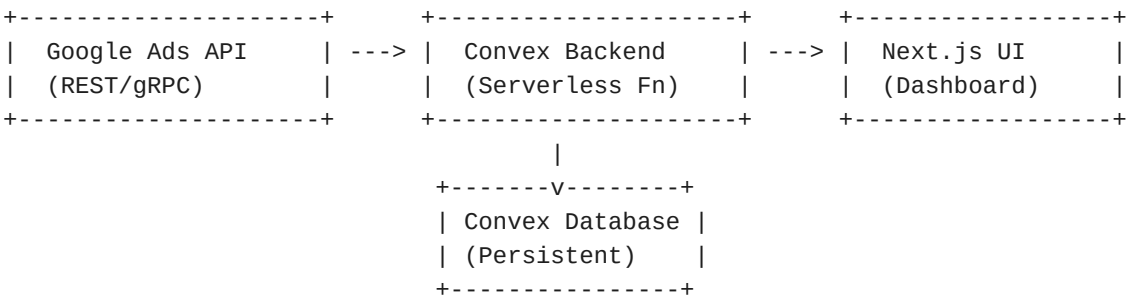
3. Architecture & Data Flow

3.1 System Architecture

The dashboard follows a simple three-tier architecture:

1. Data Ingestion Layer: Scheduled Convex actions (cron jobs) that call the Google Ads API daily to fetch the latest performance data.
2. Data Storage Layer: Convex database stores normalized campaign, ad group, and keyword data with upsert logic to prevent duplicates.
3. Presentation Layer: Next.js frontend renders charts, tables, and KPI cards from the stored data using Recharts for visualization.

3.2 Data Flow Diagram



3.3 Data Flow Steps

- Step 1: A daily cron job (06:00 UTC) triggers the syncGoogleAds Convex action.
- Step 2: The action authenticates via OAuth 2.0 using a service account or refresh token.
- Step 3: The action calls the Google Ads API (GoogleAdsService.SearchStream) to fetch metrics.
- Step 4: Results are normalized and upserted into the Convex database (campaigns, ad groups, keywords).
- Step 5: The Next.js frontend queries the Convex database in real-time and renders dashboards.

4. Google Ads API Usage

4.1 API Methods Used

The tool uses exclusively READ-ONLY operations. No campaigns, ads, or settings are created or modified through the API.

API Service	Method	Purpose
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GoogleAdsService	SearchStream	Fetch campaign metrics
GoogleAdsService	SearchStream	Fetch ad group metrics
GoogleAdsService	SearchStream	Fetch keyword metrics
CustomerService	ListAccessible	List managed accounts

4.2 GAQL Queries

Example query for campaign performance data:

```
SELECT campaign.name, campaign.status,
       campaign.advertising_channel_type,
       campaign_budget.amount_micros,
       metrics.clicks, metrics.impressions,
       metrics.cost_micros, metrics.conversions
FROM campaign
WHERE segments.date DURING LAST_30_DAYS
      AND campaign.status != "REMOVED"
```

Example query for keyword performance data:

```
SELECT ad_group.name, ad_group_criterion.keyword.text,
       ad_group_criterion.keyword.match_type,
       ad_group_criterion.quality_info.quality_score,
       metrics.clicks, metrics.impressions,
       metrics.cost_micros, metrics.conversions,
       metrics.average_cpc
FROM keyword_view
WHERE segments.date DURING LAST_30_DAYS
```

5. Supported Campaign Types

The dashboard imports and displays data from the following Google Ads campaign types:

Campaign Type	API Enum	Usage
Search	SEARCH	Lead generation (primary)
Performance Max	PERFORMANCE_MAX	Automated lead gen
Display	DISPLAY	Brand awareness
Video	VIDEO	YouTube campaigns

6. Authentication & Security

6.1 OAuth 2.0 Authentication

The tool authenticates with the Google Ads API using OAuth 2.0 with a refresh token. The authentication flow is:

1. A one-time OAuth consent flow is completed by an administrator to obtain a refresh token.
2. The refresh token is stored as an encrypted environment variable in the Convex deployment.
3. At runtime, the Convex action uses the refresh token to obtain short-lived access tokens.
4. Access tokens are used for API calls and are never stored persistently.

6.2 Credential Storage

- Developer token: Stored as encrypted environment variable (GOOGLE_ADS_DEVELOPER_TOKEN)
- Client ID/Secret: Stored as encrypted environment variables
- Refresh token: Stored as encrypted environment variable (GOOGLE_ADS_REFRESH_TOKEN)
- No credentials are hardcoded in source code
- No credentials are logged or exposed to the frontend

6.3 Access Control

- Dashboard access is restricted to authenticated NetCo employees only
- API credentials are accessible only to the backend (Convex serverless functions)
- The frontend never has direct access to API tokens or credentials

7. Rate Limiting & Compliance

7.1 API Call Volume

The tool makes minimal API calls:

- Daily sync: ~3-5 API calls per brand (3 brands = ~9-15 calls/day)
- No real-time API calls from the user interface
- All data is cached in the Convex database after import
- Estimated monthly volume: < 500 API calls total

7.2 Rate Limiting Implementation

- Sequential processing: API calls are made one brand at a time, never in parallel
- Daily schedule: Sync runs once per day at 06:00 UTC via cron job
- Error handling: Failed calls are logged and retried on the next daily cycle
- No retry loops or aggressive polling

8. User Interface

The dashboard presents Google Ads data in a structured, read-only interface organized into sections:

8.1 Campaign Overview

- KPI cards showing total spend, clicks, impressions, and conversions
- Campaign table with budget, spend, CTR, and cost per conversion
- Sortable by performance metrics

8.2 Ad Group Analysis

- Performance table grouped by campaign
- Filterable by active/paused status
- CTR and conversion metrics per ad group

8.3 Keyword Intelligence

- Top keywords ranked by conversions and clicks
- Quality Score distribution chart (1-10)

- Average Quality Score KPI card
- Match type badges (Broad, Phrase, Exact)

8.4 Global Filters

The dashboard provides global filter controls in the header bar that apply across all pages and tabs:

- Date range picker with presets (Last 7/14/30/90 days, current year, previous years, all time)
- Granularity toggle (Daily / Weekly / Monthly)
- Brand selector for switching between NetCo Body-Cam, BauTV+, and Microvista

Filters persist across page navigation via URL parameters.

9. Data Storage & Retention

9.1 Database Schema

Google Ads data is stored in three Convex database tables:

Table	Key Fields	Description
gadsCampaigns	brandId, period, campaign	Campaign-level metrics
gadsAdGroups	brandId, period, adGroup	Ad group-level metrics
gadsKeywords	brandId, period, keyword	Keyword-level metrics

9.2 Data Retention

- Data is retained indefinitely for historical trend analysis
- Upsert logic prevents duplicate entries (same brand + period + entity = update)
- No personally identifiable information (PII) is stored
- No user-level or click-level data is stored - only aggregated metrics

10. Terms of Service Compliance

- READ-ONLY access: The tool only reads performance data. No campaigns, ads, keywords, or settings are created, modified, or deleted via the API.
- No data reselling: All data is used exclusively for internal business analysis.
- No PII collection: The tool does not access or store any end-user data, search queries, or click-level information.
- Minimal API usage: Daily sync with < 500 calls/month, well within rate limits.
- Secure credential storage: All API credentials are stored as encrypted environment variables.
- Single MCC account: The tool only accesses accounts under MCC 134-864-7021.
- Compliance monitoring: The development team monitors API deprecation notices and updates accordingly.